GolfIt

Group 4: Sultans of Swing
Members: Anton Bilokon, Constantin Chau, Tamer Erata, Demian Piodi

Milestone 4 — April 7, 2026

1. Game Vision

GolfIt is a real-time multiplayer game that combines the precision of classic minigolf with a dynamic, interactive environment. Unlike traditional turn-based golf, players compete simultaneously in a race against each other to finish the course in the fewest strokes possible.

In the next version we will implement interactive map environment. Players won't just focus on their own ball; they can trigger environmental changes (like create bridges or closing gates) that affect their opponents' paths. This creates a strategic layer where "sabotaging" a faster player becomes a viable tactic to stay in the lead.

2. Win Condition

- **Primary Goal:** Sink the golf ball into the hole using the lowest number of strokes (moves) possible.
- **Win Conditions:** The player who has stroke the ball the least amout of times to reach the hole wins.

3. Game Rules

1. **Setup:** To start a game, a player has to create a lobby (press "Create Lobby") and wait for as many other players wished (tested smoothly with up to 4 players, lags with 5 and 6 players). To start a game one of any player in the lobby has to press the green button "START" All players spawn simultaneously at the starting tee. Each player controls a ball that at the current version is blue.

2. **Gameplay:** The only game move each player can make is to stroke a golf ball. It is done by pressing in any point of the map and drag the mause in the opposite of the wanted direction. The longest drag press with the mouse, the more powerful the stroke of the ball. There is no waiting for opponents; currently players can kick their ball also while it is moving. Ball positions and movement animations are synchronized across all clients via the server. The game operates in real time.

3. **Constraints:** Balls cannot pass through solid walls. Contact with walls and trees results in a physical bounce based on the angle of incidence. Touching the water (blue rectangles) resets the ball to its last stable position. Touching the black hole makes you finish the game and wait for the other players.

4. **Friction:** There are different terrains that slows down the ball less or more. The grass (green) has a normal friction value, the sand (brown) has a much greater friction, meaning the ball will slow down much faster, while on the ice (light blue) the ball will keep longer its velocity.

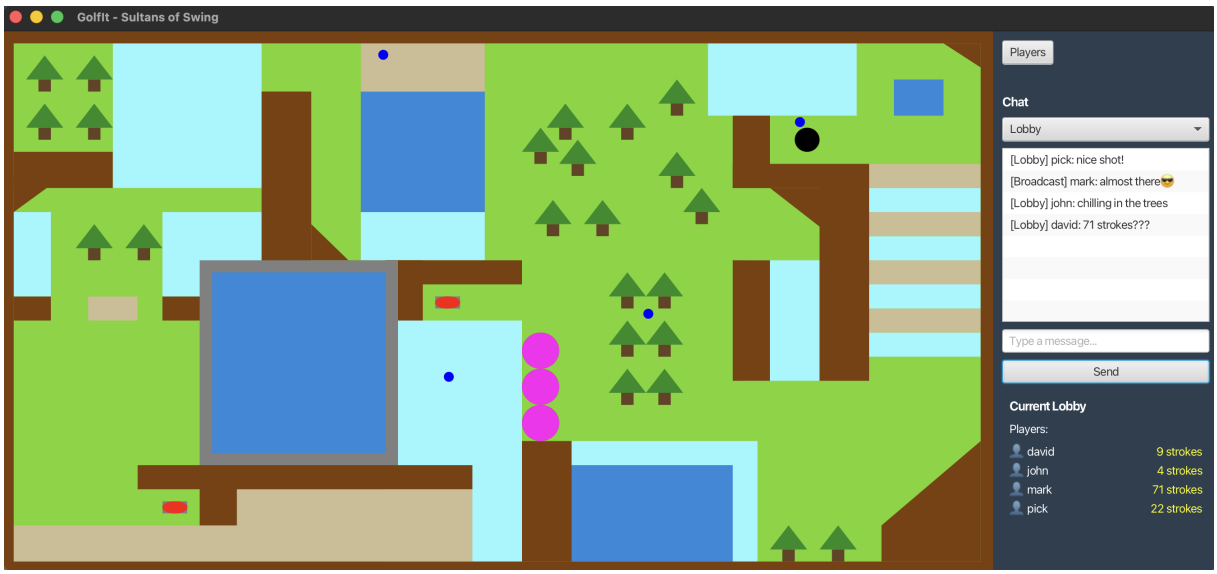


Figure 1: Gameplay view